DETECTION OF THE LIQUID LEVEL IN AN OPAQUE BOTTLE

INTRODUCTION

THE CUSTOMER REQUESTS INSPECTION OF PRODUCT FILLING IN OPAQUE PLASTIC CONTAINERS. THE PRODUCT IS A LIQUID. THE CONTAINERS ARE MADE OF PLASTIC COLORED WITH A YELLOW PIGMENT. THE LIQUID LEVEL IN THE CONTAINER IS CHECKED AND SHOULD BE 10±1CM. IF THE CONTAINER CONTAINS THE PRESCRIBED AMOUNT OF PRODUCT, THE PROGRAM RETURNS "PASSED"; OTHERWISE, IT RETURNS "FAILED."

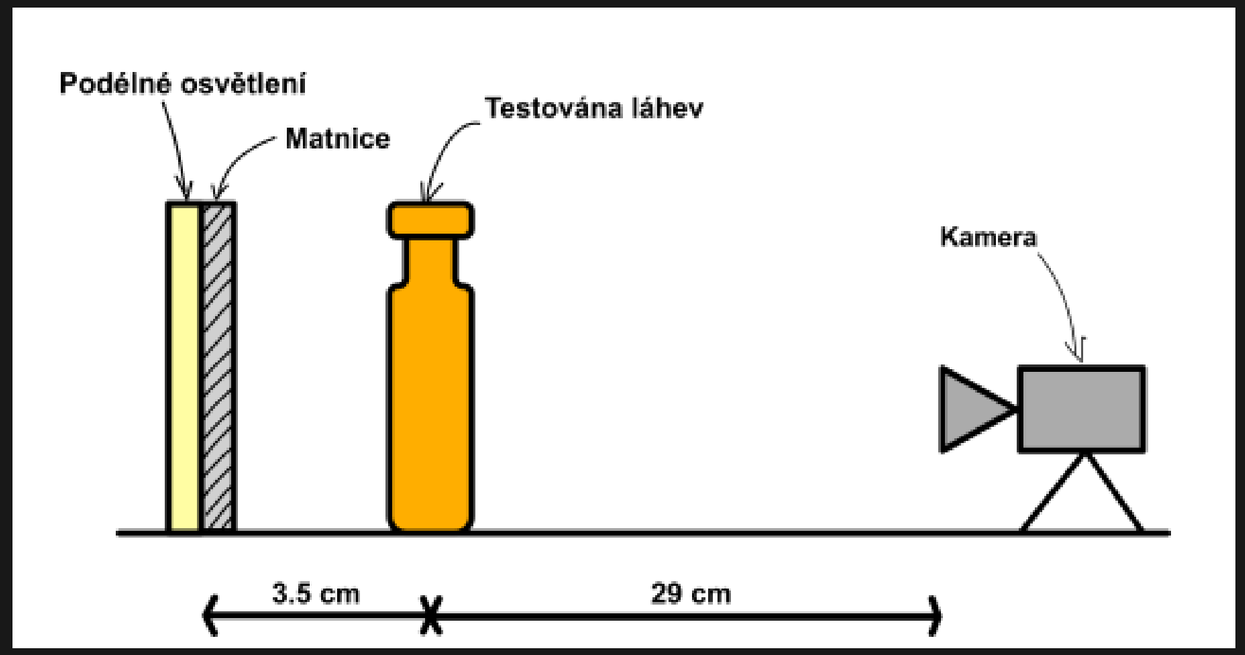
SETTING THE SCENE

- LIQUID LEVEL INSPECTION IN OPAQUE BOTTLES – LIGHTING AND IMAGING SETUP
  - GOAL: DETECT LIQUID LEVEL; SILHOUETTES AND CONTRAST BETWEEN FILLED/EMPTY PARTS ARE SUFFICIENT
  - LIGHTING: BACKSTRIP ILLUMINATION
  - BACKGROUND: BLACK PAPER TO MINIMIZE REFLECTIONS

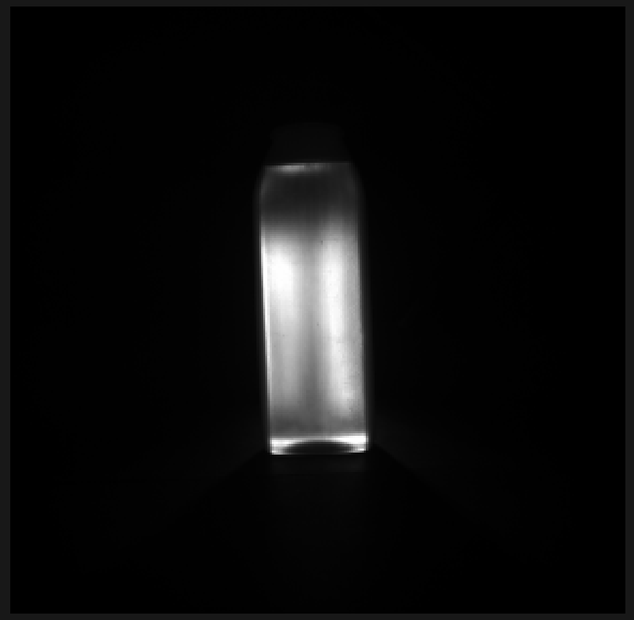
- INFRARED (IR) LIGHTING
  - SOURCE: SMARTVIEW LL-130IR50, 850 NM
  - CAMERA: BASLER ACA2040-90UMNIR (MONOCHROME)
  - ISSUES: LOW EXPOSURE → DARK BOTTOM; HIGH EXPOSURE → REFLECTIONS IN ALL DIRECTIONS

- BLUE LIGHT (RECOMMENDED)
  - SOURCE: SMARTVIEW LL-130B120, 525 NM (COMPLEMENTARY TO ORANGE)
  - CAMERA: BASLER ACE2500-60UM, MONOCHROME
  - FILTER: BP465-70HT (BANDPASS 525 NM)
  - ADVANTAGES: HIGH CONTRAST BETWEEN BOTTLE AND CONTENTS; MINIMAL AMBIENT INFLUENCE
  - POSITIONING: OBJECT 29 CM FROM CAMERA, LIGHT 3.5 CM FROM OBJECT
  - DIFFUSER: ENSURES UNIFORM ILLUMINATION AND HIGHLIGHTS EDGES

SCENE


Podélné osvětlení
Matnice
Testována láhev
Kamera
3.5 cm
29 cm

PICTURES



IR EX: 500

IR EX: 25000

BLUE LIGHT

FINAL SETTINGS

Typ kamerového systému	Basler ace2500-60um
Objektiv	$f = 12$ mm, F1.8
Nastavení kamery	ex: 3500, clona: 4
Osvětlení	Smartview LL-130B120, 525 nm, zadní pruhové
Difuzor	propustnost: 50%
Filtr	BP465-70HT
Napájecí zdroj	USB 3.0 Micro-B connector
Kontrolér	Basler Pylon
Rozměry kontrolovaného dílu	v×š×h: $185 \times 60 \times 40$ [mm]
Barvy dílů	lahev: oranžový plast, tekutina: bezbarvý šampón
Díly v pohybu/stacionární	stacionární
Prostředí	Výrobní pás
Okolní osvětlení	Denní světlo
Komunikační rozhraní kamery	USB 3.0 Micro-B connector

IMAGE PROCESSING

USED FUNCTIONS, LOGIC, RESULTS, WHAT WORKED AND WHAT DIDNT, SCREEN OF PROGRAM (JUST THE TABLE OR ALL USED SETTINGS ARE ENOUGHT)

CONCLUSIONS